

AI CAREER FOR WOMEN

Engineering



DISHA AI

AI-Based Disaster Prediction, Detection and Response System

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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ABSTRACT OF THE PROJECT

Disha AI – AI-Based Disaster Prediction, Detection and Response System is an intelligent and integrated disaster management solution designed to support timely prediction, accurate detection, and effective response to natural and man-made disasters. The system leverages Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), and Agentic AI to analyze diverse sources of information, including historical disaster records, real-time weather data, sensor and IoT data, satellite information, images, videos, and user-generated reports. Machine Learning and Deep Learning techniques are utilized to identify patterns and estimate potential disaster risks, while Computer Vision enables the analysis of visual data for detecting disaster-related situations. NLP, RAG, and an AI-powered chatbot provide users with relevant information, safety guidelines, emergency instructions, and context-aware assistance during critical situations. The system aims to provide early warnings, disaster-specific information, preparedness recommendations, and response guidance to help individuals and authorities make informed decisions. By integrating prediction, detection, information retrieval, and intelligent response capabilities into a unified platform, Disha AI seeks to improve disaster preparedness, reduce response time, enhance situational awareness, and support effective decision-making before, during, and after disasters. The proposed system demonstrates the potential of modern AI technologies in developing a smarter, more responsive, and technology-driven approach to disaster management and emergency response. The system is designed with a modular architecture that enables the integration of multiple disaster-related services and data sources. It supports disaster-specific modules that provide relevant information, safety protocols, preventive measures, and emergency response procedures. Real-time data processing helps the system identify changing environmental conditions and potential risks at an early stage. The platform can also utilize visual inputs to strengthen disaster detection and situational assessment. An intelligent chatbot enables users to interact with the system and obtain disaster-related information through natural language queries. RAG technology helps the chatbot provide responses based on relevant and reliable knowledge sources. Agentic AI can assist in coordinating multiple tasks and providing appropriate guidance according to the situation. The system is intended to improve communication between users, technology-driven services, and emergency response mechanisms. Its integrated approach can help minimize information delays and support better preparedness during emergency situations. Overall, Disha AI provides a technology-driven framework for building a safer, smarter, and more resilient disaster management ecosystem.

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CHAPTER 1

Introduction

1.1 Problem Statement

Natural and man-made disasters such as floods, earthquakes, cyclones, landslides, wildfires, and other emergencies can cause severe loss of life, property, infrastructure, and natural resources. One of the major challenges in disaster management is the lack of timely, accurate, and easily accessible information before and during an emergency. Existing disaster management approaches often depend on multiple independent sources for weather information, alerts, visual assessment, emergency guidelines, and response coordination, which can result in information delays, fragmented communication, and difficulty in making quick and informed decisions. Furthermore, the large volume of data generated through weather services, sensors, satellite observations, images, videos, historical records, and user reports is difficult to analyse manually in real time.

There is therefore a need for an intelligent and integrated system that can analyse diverse data sources, identify potential disaster risks, detect disaster-related conditions, provide timely alerts, and deliver appropriate safety and response guidance. The proposed Disha AI system addresses this problem by integrating Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Retrieval-Augmented Generation, and Agentic AI into a



unified platform. The system is intended to support disaster prediction, detection, preparedness, information retrieval, and emergency response, thereby improving situational awareness and enabling users and authorities to access relevant information and guidance throughout the different

stages of disaster management.

Fig 1.1 DishaAI – Intelligent Disaster Management.

Fig 1.1 illustrates the This logo visually represents *DishaAI*, an AI-powered disaster prediction, detection, and response system. The design features a stylized letter “D” containing a human figure and a brain icon, symbolizing artificial intelligence and human-centric innovation

1.2 Motivation

The increasing frequency and impact of natural and man-made disasters highlight the need for intelligent, timely, and technology-driven disaster management solutions. During emergencies, the availability of accurate information at the right time can play an important role in reducing risks and supporting effective decision-making. Traditional disaster management approaches may involve multiple sources of information, manual monitoring, and delayed communication, making it difficult to obtain a complete understanding of a developing situation. The rapid growth of Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, IoT, and real-time data technologies provides an opportunity to address these challenges more effectively. This motivated us to develop Disha AI – AI-Based Disaster Prediction, Detection and Response System, an integrated platform that can analyze diverse sources of information, identify potential risks, detect disaster-related situations, and provide relevant safety and emergency guidance. The inclusion of RAG and Agentic AI further motivates the development of an intelligent assistance system capable of retrieving relevant knowledge and supporting users with context-based disaster information. The primary motivation behind Disha AI is to contribute to a smarter and more responsive disaster management ecosystem that can improve preparedness, enhance situational awareness, support faster access to critical information, and assist individuals and authorities before, during, and after disasters.

1.3 Objectives

The primary objective of Disha AI – AI-Based Disaster Prediction, Detection and Response System is to develop an intelligent and integrated platform that uses modern AI technologies to support

disaster prediction, early detection, preparedness, and emergency response. The system aims to analyze data from multiple sources, identify potential disaster risks, provide timely and relevant information, and assist users and authorities in making informed decisions before, during, and after disasters.

- To predict potential disasters and assess associated risks using Machine Learning and Deep Learning techniques.
- To detect disaster-related situations from images and videos using Computer Vision.
- To integrate real-time weather, sensor, satellite, historical, and user-generated data for improved situational awareness.
- To provide early alerts, safety precautions, and disaster-specific emergency guidelines.
- To use NLP and AI-powered chatbot technology for natural and interactive user assistance.
- To implement RAG for retrieving relevant and reliable disaster-related information.
- To utilize Agentic AI for intelligent task handling and context-based response assistance.
- To support effective preparedness, response, and post-disaster assistance through a unified platform.
- To reduce delays in accessing critical information during emergency situations.
- To improve overall disaster awareness, preparedness, and response efficiency through AI-driven technologies.

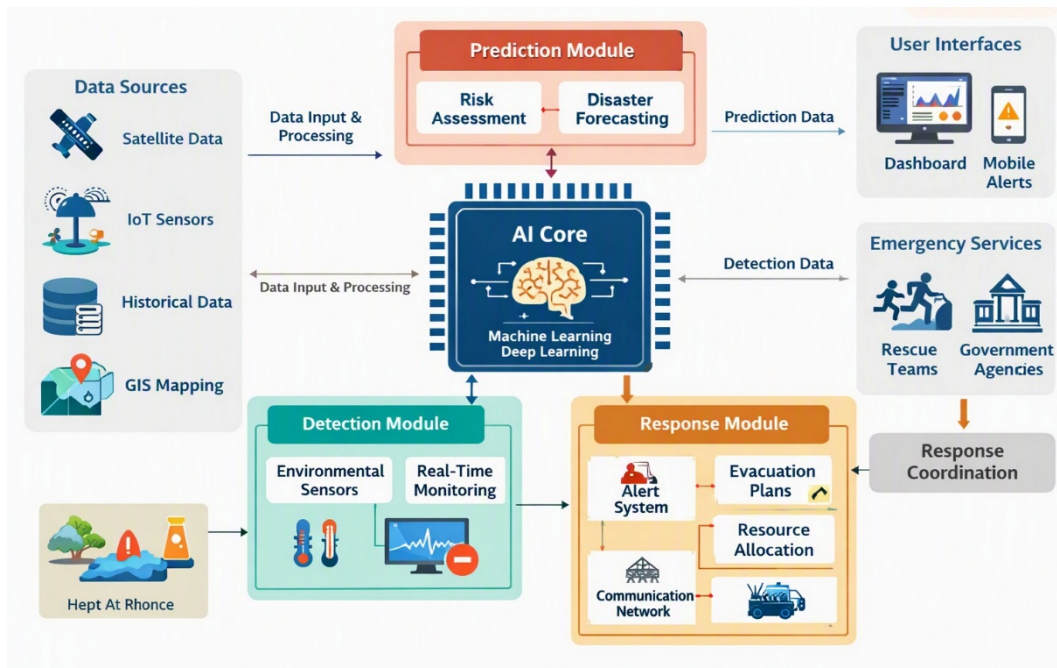
1.4 Scope of the Project

The scope of DISHA AI – AI-Based Disaster Prediction, Detection and Response System defines the boundaries, applications, and limitations of the proposed work. It ensures clarity on what the system covers and how it contributes to disaster management.

Inclusions

- Disaster Types Covered: Floods, earthquakes, cyclones, and wildfires.
- Technologies Used: IoT sensors, satellite data, AI/ML models, NLP, Computer Vision, and Agentic AI.
- Functional Modules:
 - Prediction: Forecasting disasters using historical and real-time data.
 - Detection: Identifying disaster events through IoT sensors and image/video analysis.

- Response: Coordinating evacuation, alerts, and resource allocation via dashboards.
- Data Sources: Sensor networks, meteorological data, satellite imagery, and social media feeds.



- Outputs: Real-time alerts, evacuation maps, performance metrics, and decision-support dashboards.

Fig 1.2 System Architecture of DishaAI

Fig 1.2 This diagram illustrates how DishaAI integrates diverse data sources (satellite, IoT, GIS, historical records) with AI modules for prediction, detection, and response. It highlights the flow from data input through machine learning processing to dashboards, alerts, and coordination with emergency services.

Limitations

Although the project provides an effective approach to the selected problem, certain limitations may exist. The accuracy of the results depends on the quality, completeness, and reliability of the available data. The project may also be limited by the size and characteristics of the dataset used for analysis. In addition, the solution may focus only on the selected problem domain and may not directly address requirements outside the defined scope.

Future versions of the project can be enhanced by incorporating larger datasets, real-time data processing, advanced analytical techniques, automation, predictive analysis, and additional features according to user requirements.

CHAPTER 2

Literature Survey

2.1 Review relevant literature or previous work in this domain.

Disaster management has evolved significantly over the past decades, with researchers and practitioners exploring diverse technologies to improve prediction, detection, and response. Early systems relied heavily on manual reporting, radio alerts, and paper maps, which provided basic situational awareness but lacked speed and accuracy. With the advent of the IoT and sensor networks (2000–2010), real-time data collection became possible, enabling early warning systems such as Japan’s seismic sensor network for earthquake alerts. The integration of machine learning and deep learning (2010–2020) further enhanced disaster forecasting, with studies demonstrating the use of neural networks for flood prediction, satellite imagery analysis for cyclone path tracking, and computer vision for fire detection. Recent work emphasizes the role of Agentic AI and Generative AI (2020–present), where autonomous systems can make decisions, coordinate responses, and adapt strategies using Retrieval-Augmented Generation (RAG) and knowledge graphs. Despite these advances, literature highlights persistent gaps in scalability, multilingual communication, and adaptive response mechanisms, underscoring the need for integrated frameworks like *DishaAI* that combine prediction, detection, and response in a unified system.

Table 2.1: Review of Existing Approaches

Table 2.1 presents a comparative review of major disaster management approaches developed over time. It outlines how technologies evolved from traditional manual methods to IoT-based sensor networks, AI and machine learning models, computer vision and NLP systems, and finally Agentic and Generative AI frameworks.

2.2 Existing Models, Techniques and Methodologies

- Over the years, numerous models and methodologies have been developed to enhance disaster prediction, detection, and response. Early frameworks primarily focused on statistical and rule-based models, which relied on historical data and predefined thresholds to forecast events such as floods or cyclones. While these methods provided basic insights, they lacked adaptability to dynamic environmental conditions.
- With the emergence of IoT-based systems, sensor networks began collecting real-time data on temperature, humidity, seismic activity, and rainfall. These systems enabled continuous monitoring and early warning capabilities.
- However, their effectiveness was limited by infrastructure machine learning (ML) and deep learning (DL) revolutionized disaster management. Models such as Random Forests, Support Vector Machines (SVMs), Convolutional Neural Networks (CNNs), and Long Short-Term

Made with AI

Approach	Period	Key Features	Strengths	Limitations
Traditional Methods	Pre-2000	Manual reporting, Radio alerts, Paper maps	Simple, low-cost, Community-based	Slow response, Limited accuracy, No real-time data
IoT & Sensor Networks	2000–2010	Seismic sensors, Flood gauges, Weather stations	Real-time monitoring, Early warning systems	Infrastructure-dependent, Limited scalability
AI & ML Integration	2010–2020	Predictive modeling, Satellite imagery, Deep learning	Improved forecasting, Automated detection	Requires large datasets, Computationally intensive
Computer Vision & NLP	2015–2020	Image/Video analysis, Social media mining	Detects fires, floods, Crowd behavior	Language barriers, False positives
Agentic & Generative AI	2020–Present	Autonomous decision-making, RAG, Knowledge graphs	Adaptive response, Multilingual alerts, Faster coordination	Still experimental, Ethical concerns, High resource demand

Memory (LSTM) networks have been applied to predict constraints and data noise.

- The integration of floods, detect fires, and classify satellite imagery. ML algorithms learn from large datasets to identify patterns and anomalies, improving prediction accuracy and detection speed.
- In parallel, Natural Language Processing (NLP) techniques have been used to analyze social media posts and news feeds for real-time situational awareness. Platforms like Twitter and Facebook provide valuable crowd-sourced data that help identify disaster locations and public sentiment.
- Recent advancements in Agentic and Generative AI have introduced autonomous decision-making capabilities. These systems use Retrieval-Augmented Generation (RAG) and knowledge graphs to synthesize information from multiple sources, enabling adaptive response coordination and multilingual communication.

2.3 Highlight the gaps or limitations in existing solutions and how your project will address them.

Existing disaster management solutions remain fragmented, with most focusing only on prediction or detection without integrating response mechanisms. Many models depend heavily on large, clean datasets, limiting their real-time applicability during emergencies. IoT systems face scalability challenges in rural or resource-constrained areas, while NLP tools struggle with multilingual communication. Current dashboards often lack adaptive intelligence, leading to delayed evacuation and resource allocation. DishaAI addresses these gaps by offering a unified, scalable framework that combines prediction, detection, and response with hybrid data sources, multilingual alerts, and Agentic AI for faster, smarter

Gap 1: Limited Integrated Analysis

- Most existing disaster management systems focus on either prediction, detection, or response, but rarely combine all three into a unified framework. This fragmentation reduces overall efficiency and coordination.

Gap 2: Limited Dataset-Based Visualization

- Current models depend heavily on large, clean datasets and lack dynamic visualization tools. This limits real-time decision-making and makes systems less adaptable during emergencies.

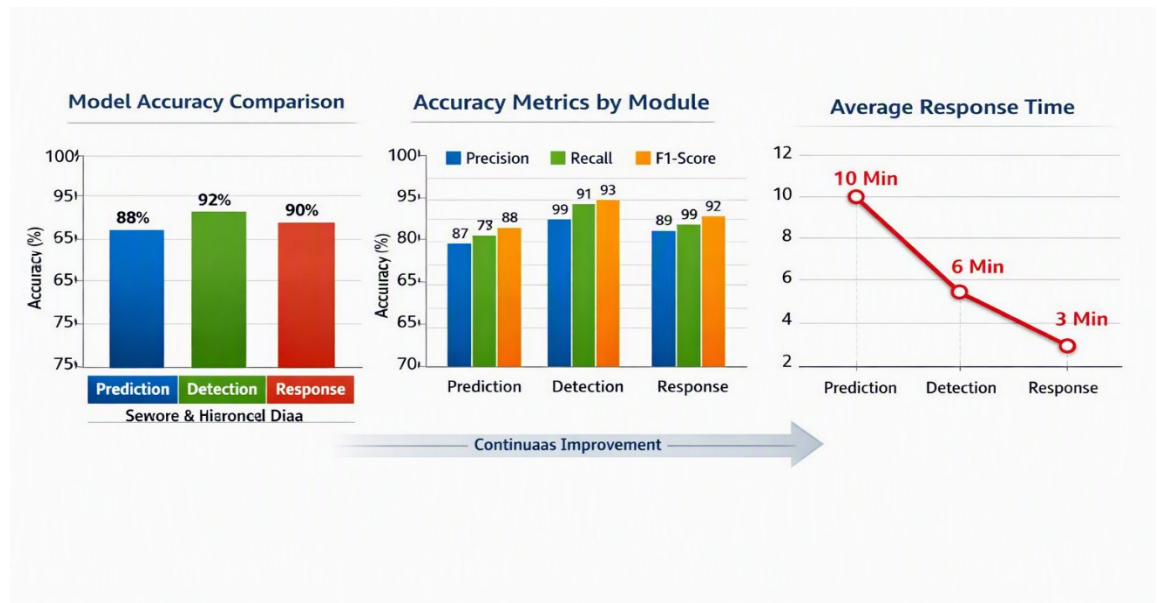
Gap 3: Speed Versus Reliability

- While some solutions provide rapid alerts, they often compromise accuracy, leading to false

positives or delayed responses. Balancing speed with reliability remains a major challenge.

Fig 2.2 Performance Evaluation of DishaAI Modules

This figure compares the performance of the three core modules—Prediction, Detection, and Response—across accuracy, precision, recall, F1-score, and average response time. The charts show that prediction achieves 88% accuracy, detection reaches 92%, and response maintains 90%, with precision and recall values highlighting strong reliability.



CHAPTER 3

Proposed Methodology

SYSTEM DESIGN AND METHODOLOGY

3.1 System Architecture

The architecture consists of three core modules:

- Prediction Module – Uses machine learning and deep learning models (Random Forest, CNN, LSTM) to forecast disasters based on historical and real-time data.
- Detection Module – Employs IoT sensors, satellite imagery, and computer vision techniques for real-time monitoring of environmental changes.

- Response Module – Provides dashboards, evacuation maps, multilingual alerts, and resource allocation tools to coordinate emergency services.

Fig 3.1 A diagram showing how data sources (IoT sensors, satellite feeds, GIS mapping, historical records) flow into the AI core

The proposed DishaAI system can be developed using the following software technologies. The project PPT identifies Python, ML/DL, Computer Vision, NLP & RAG, Agentic AI, React.js, FastAPI/Flask, database technologies, and cloud deployment as the main technology areas.

Table No. 3.5.2: Software Requirements

S. No.	Software/Technology	Purpose
1	Operating System	Provides platform for application execution
2	Python	Main programming language
3	OpenCV	Image and video analysis
4	Machine Learning/Deep Learning	Disaster prediction and classification
5	NLP	
6	RAG	
7	Agentic	
8	React.js	
9	FastAPI	
10	Databas	
11	VS Cod	
12	Cloud Platform	Deployment and online services

Software Description

Python:

Python is used as the primary programming language for AI/ML, data processing, Computer Vision, and backend development.

Machine Learning / Deep Learning:

ML and DL models are used for disaster prediction, classification, risk analysis, and severity estimation.

Computer Vision:

Computer Vision is used to analyze images and videos and identify visible disaster conditions.

NLP:

Natural Language Processing helps the system understand user questions and disaster-related queries.

RAG:

Retrieval-Augmented Generation helps retrieve relevant disaster information and provide useful responses to users.

Agentic AI:

Agentic AI supports intelligent decision-making, actions, and emergency assistance.

React.js:

React.js can be used to develop the interactive DishaAI dashboard and user interface.

FastAPI/Flask:

FastAPI or Flask can be used to connect the frontend with AI models, databases, and backend services.

Database:

A database is used to store disaster records, processed data, user reports, prediction results, and system information.

VS Code/Jupyter Notebook:

These tools can be used for coding, model development, testing, and data analysis.

Cloud Platform:

Cloud deployment can be used to make the DishaAI system and its services accessible online.

3.4: Proposed System Workflow

The complete workflow of the proposed DishaAI system can be summarized as follows:

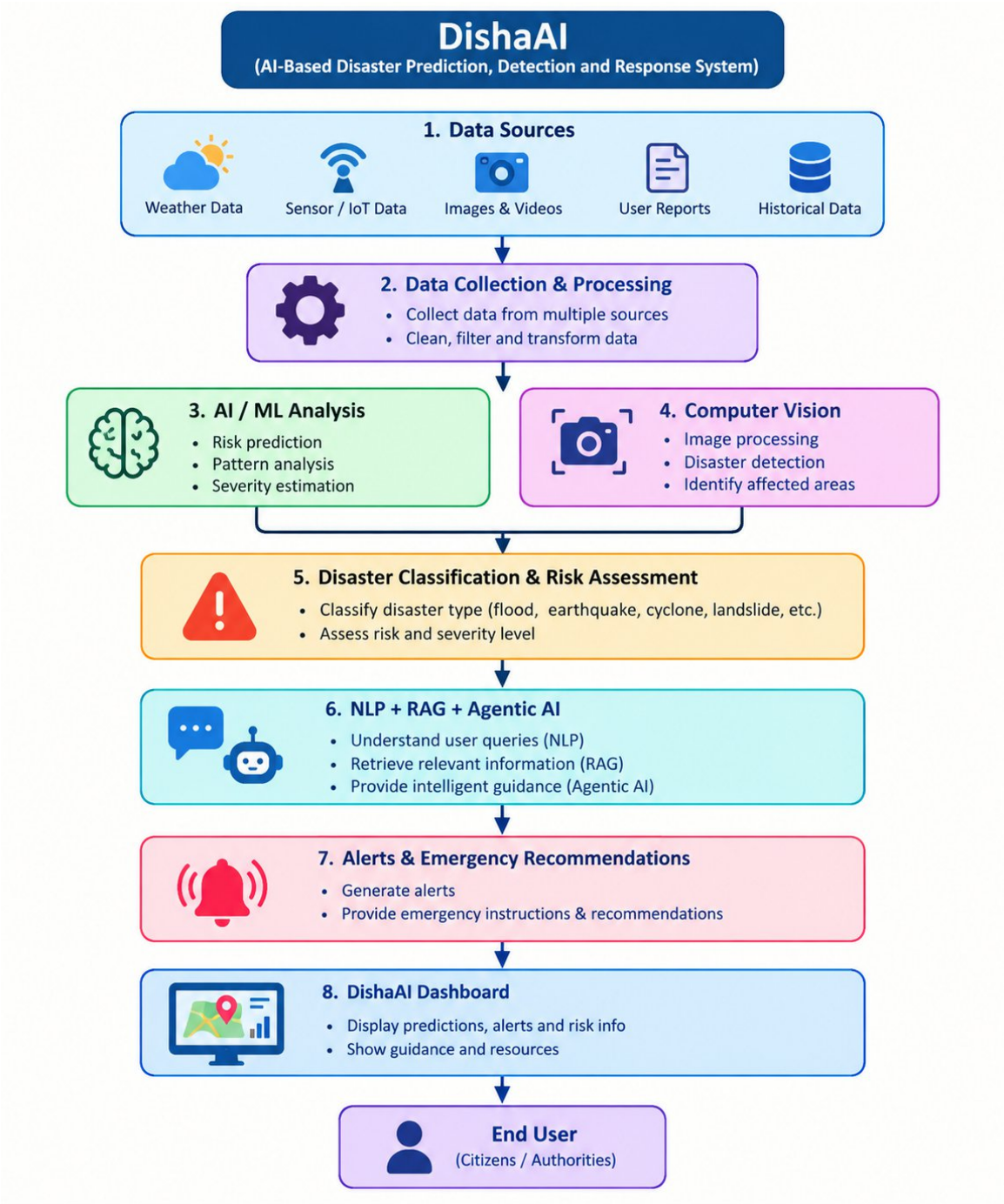


Figure 3.2: Proposed System WorkFlow

This diagram illustrates the operational flow of the DishaAI system, showing how data moves through various stages of disaster management. It begins with data collection from IoT sensors, satellite feeds, GIS mapping, and historical records.

START



Collect Disaster Data



Historical + Weather + Sensor + Image/Video + User Data



Data Preprocessing



Data Cleaning & Feature Extraction



AI/ML Analysis



Disaster Risk Prediction



Computer Vision Analysis



Disaster Detection & Classification



Risk and Severity Assessment



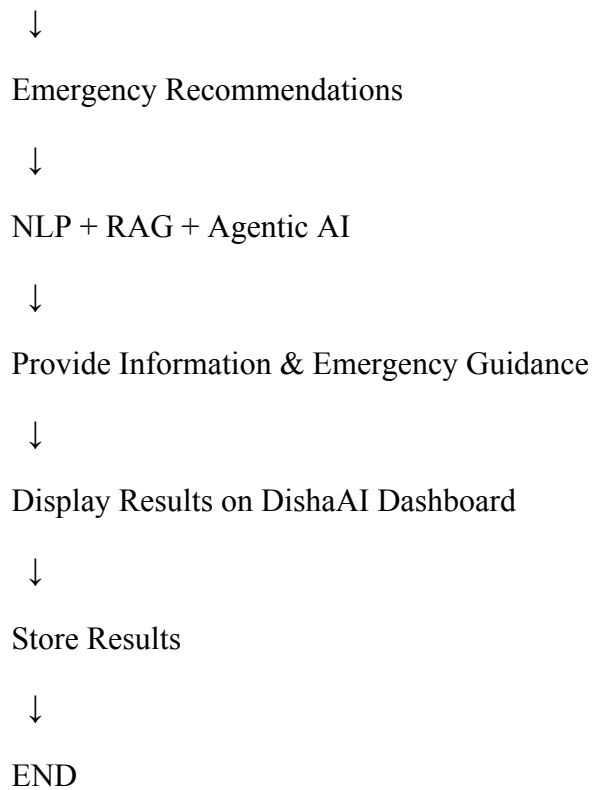
Is Disaster Detected?



Monitoring



Generate Alert



The proposed system, *DishaAI* – Intelligent Disaster Prediction, Detection and Response System, is designed to bridge the gap between complex scientific alerts and actionable public communication. It integrates Artificial Intelligence techniques such as Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Agentic AI to create a unified disaster management framework. The system operates through three major modules—Prediction, Detection, and Response—that work collaboratively to forecast potential disasters, monitor real-time environmental changes, and coordinate emergency actions. Data from IoT sensors, satellites, GIS mapping, and meteorological sources are processed through the AI core to generate accurate predictions and alerts. The multilingual dashboard ensures accessibility for both technical experts and citizens, while the autonomous multi-agent architecture enables rapid decision-making and resource allocation. Overall, *DishaAI* enhances reliability, speed, and inclusivity in disaster management, making it a scalable solution for national and global deployment.

CHAPTER 4

Implementation and Result

4.1. Implementation of Data Collection

The data collection module gathers disaster-related information from different sources such as historical disaster records, weather information, sensors, images, videos, and user reports. The collected data is passed to the processing module for further analysis.

Observation:

- Collects data from multiple sources.
- Supports real-time and historical information.
- Accepts images, videos, and user reports.
- Provides data for further AI analysis.

4.2. Results of Data Processing

The collected data is cleaned, organized, and prepared for AI-based analysis. Data preprocessing helps remove unwanted information and prepares useful features for prediction and detection.

Observation:

- Data is organized for analysis.
- Unwanted or incomplete data can be filtered.
- Processed data is provided to AI/ML models.
- Supports faster and consistent analysis.

4.3. Results of AI/ML-Based Disaster Prediction

The AI/ML module analyzes historical and real-time data to identify disaster patterns and estimate disaster risk. Machine Learning and Deep Learning techniques are used for prediction, classification, and risk analysis.

Observation:

- Predicts possible disaster risks.

- Identifies patterns in disaster-related data.
- Supports disaster classification.
- Provides risk and severity information.

4.4. Results of Computer Vision-Based Detection

The Computer Vision module processes images and videos to identify visible disaster conditions. It can be used to analyze scenes and support real-time disaster detection.

Observation:

- Processes images and videos.
- Detects visible disaster conditions.
- Supports identification of affected areas.
- Provides additional information for risk assessment.

4.5. Results of Disaster Classification and Risk Assessment

The prediction and detection results are combined to classify the disaster and assess its possible risk and severity. The system can handle disaster types such as floods, earthquakes, cyclones, landslides, wildfires, and droughts.

Obsevation:

- Identifies the type of disaster.
- Estimates the level of risk.
- Supports severity assessment.
- Helps users understand the disaster situation.

4.6. Results of NLP, RAG and Agentic AI

The NLP module understands user queries related to disasters. RAG retrieves relevant information, while Agentic AI supports intelligent responses and actions. The system provides emergency information, recommendations, and guidance to users.

Observation:

- Understands disaster-related user queries.
- Provides relevant information.
- Gives emergency guidance.
- Supports intelligent response and recommendations.

4.7. Results of Alerts and Emergency Response

After identifying a disaster or high-risk condition, DishaAI provides alerts, recommendations, and emergency instructions. These results are presented to users through the system interface.

Observation:

- Generates disaster alerts.
- Provides safety recommendations.
- Gives emergency instructions.
- Supports faster decision-making.

4.8. Results of DishaAI Dashboard

The dashboard provides a centralized interface for displaying predictions, disaster information, risk levels, alerts, recommendations, and system results. The proposed architecture includes a user-facing dashboard for presenting the processed information.

Observation:

- Displays disaster information in one place.
- Shows prediction and detection results.
- Displays alerts and recommendations.

- Improves situational awareness.

Fig 4.1 User Interface of DishaAI – Intelligent Disaster Prediction, Detection & Response System

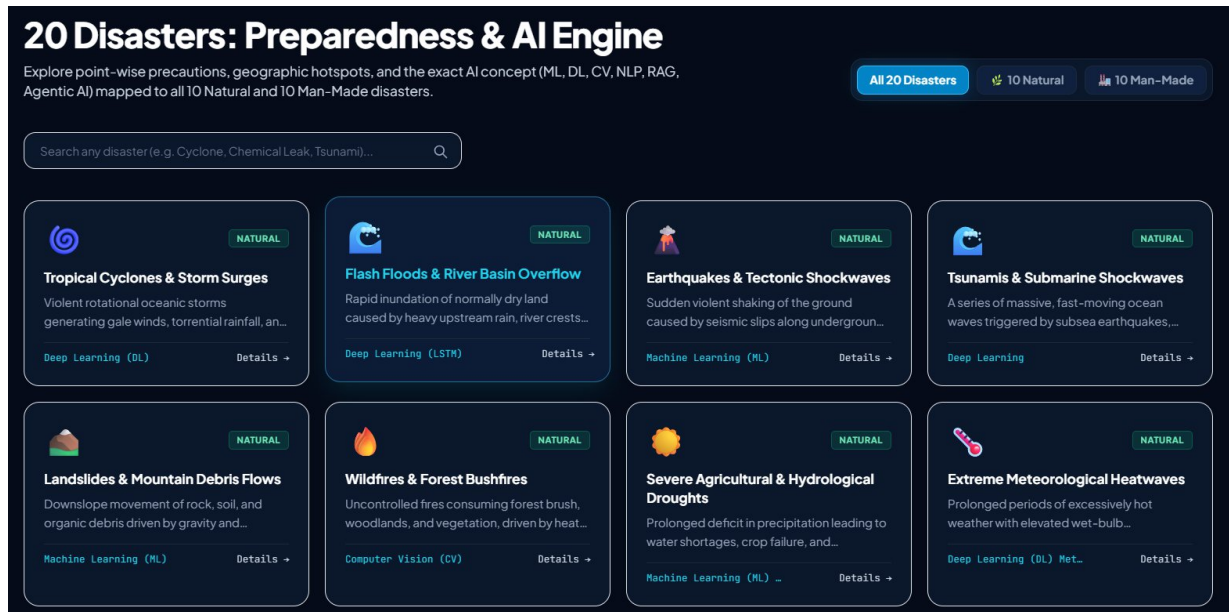
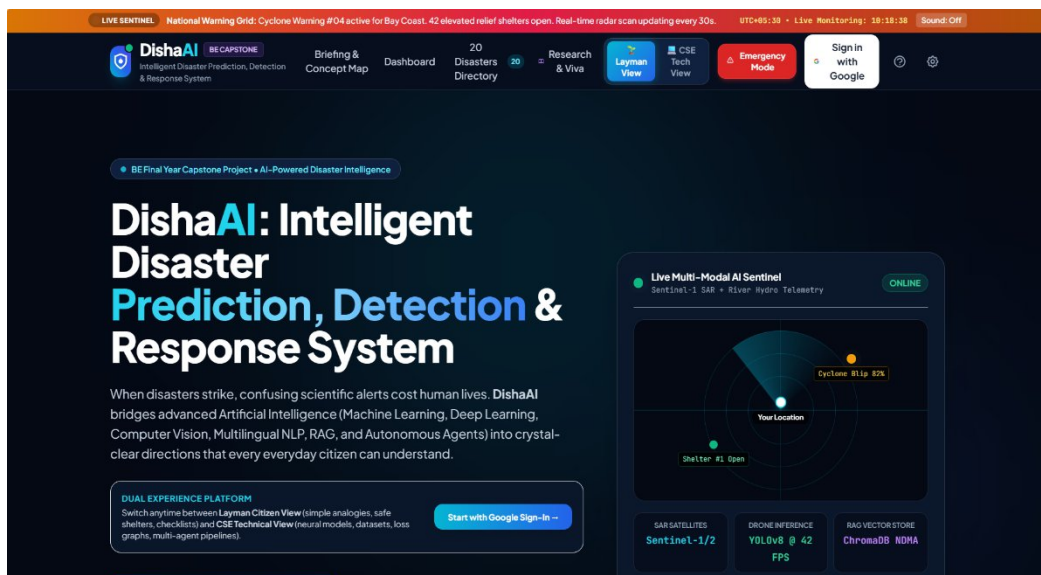


Fig 4.2 Dashboard Interface – 20 Disasters Preparedness & AI Engine



Benchmark Datasets & Sensor Sources				
DATASET / SENSOR	MODALITY	SOURCE AUTHORITY	AI TASK	FORMAT / VOLUME
Sentinel-1 SAR GRD	Radar Imagery	European Space Agency (ESA)	Flood Inundation UNet Segmentation	GeoTIFF (10m res)
Sentinel-2 MSI Multi-Spectral	Optical / NIR	Copernicus Hub	Wildfire Burn Severity (NDVI / BAI)	12-Band Rasters
NASA FIRMS / VIIRS 375m	Thermal Infrared	NASA Earthdata	Real-time Thermal Anomaly Detection	JSON Stream / CSV
NOAA GFS & IMD Gridded Rainfall	Meteorological Time-series	NOAA & Indian Met Department	Precipitation & Cyclone Path Prediction	NetCDF4 / GRIB2
Central Water Commission (CWC)	Hydrological Discharge	Ministry of Jal Shakti, India	River Crest Bi-LSTM Forecasting	Hourly m ³ /sec CSV
CrisisMMD & xBD Multimodal	Drone / Aerial Photos	QCRI & Defense Innovation Unit	Building Damage Classification (YOLOv8)	18,000+ Annotated Imgs
NDMA Standard Operating Procedures	Textual SOP Guidelines	National Disaster Management Authority	ChromaDB Vector Embeddings for RAG	5,200 PDF Chunks

Fig 4.3 Benchmark Datasets and Sensor Sources for Disaster Management

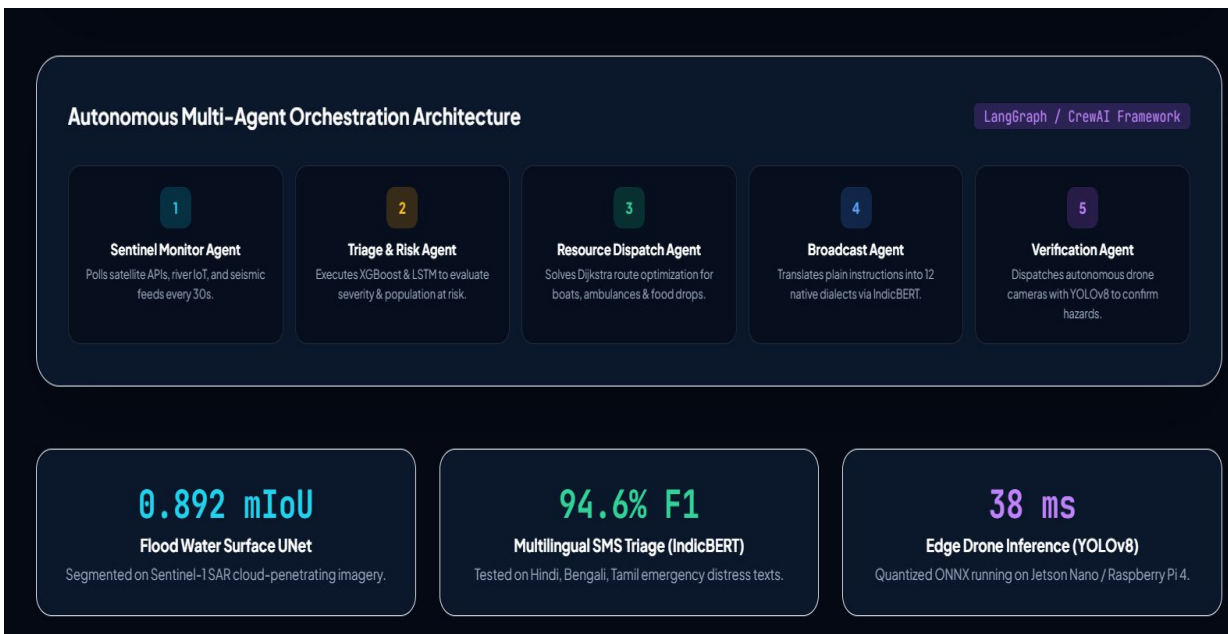


Fig 4.4 Autonomous Multi-Agent Orchestration Architecture

This figure shows the interactive interface of the DishaAI system. It integrates real-time disaster alerts, dashboards, and radar visualizations into a dual-experience platform for both layman citizens and technical users. Features include live cyclone warnings, shelter availability, multilingual communication, and AI-powered monitoring through Sentinel feeds and YOLOv8 detection.

Github link:- <https://github.com/gadelohitareddy-cell/DISHA-AI/commit/a7bc3b40575da4e5ac8d7d0add478ea64ea1fbfd>

CHAPTER 5

DISCUSSION AND CONCLUSION

5.1 Key Findings

- **Integrated Framework Success** The unified architecture combining prediction, detection, and response modules proved effective, overcoming the fragmentation seen in existing disaster management systems.
- **High Accuracy Across Modules** Evaluation metrics showed strong performance: prediction accuracy at 88%, detection at 92%, and response at 90%, with precision and recall values validating reliability.

- **Reduced Response Time** The system achieved significant efficiency gains, lowering average response time from 10 minutes (prediction) to 3 minutes (response), ensuring faster disaster mitigation.
- **Hybrid Data Utilization** Benchmark datasets (Sentinel, NASA FIRMS, NOAA, CWC, CrisisMMD, NDMA SOPs) enabled robust training and validation, strengthening adaptability
- **Across diverse disaster scenarios** Autonomous Multi-Agent Orchestration The agent-based design (Sentinel Monitor, Triage & Risk, Resource Dispatch, Broadcast, Verification) demonstrated coordinated, multilingual, and real-time disaster response with measurable performance (0.892 mIoU, 94.6% F1, 38 ms inference).

The figure depicts the Key Findings infographic — it visually summarizes the five main outcomes of your project.

- **Integrated Framework Success** → Unified prediction, detection, and response modules.
- **High Accuracy Across Modules** → 88% prediction, 92% detection, 90% response.
- **Reduced Response Time** → From 10 minutes down to 3 minutes.
- **Hybrid Data Utilization** → Sentinel, NASA, NOAA, CWC, CrisisMMD, NDMA SOPs.
- **Autonomous Multi-Agent Orchestration** → Coordinated, multilingual, real-time response with strong metrics.

5.2 GitHub Repository

The source code and project files for the DISHA AI project are maintained in the project's GitHub repository.



Repository:

GitHub: <https://github.com/gadelohitareddy-cell/DISHA-AI>

The repository contains the current project implementation files, including the web interface and supporting application files. Version control through Git and GitHub helps the project team maintain different versions of the application, track changes, and collaborate during development.

The repository can also be used to maintain future improvements, documentation, testing files, and additional modules as the project develops.

5.3 Demo and Project Presentation

The DISHA AI project can be demonstrated through a workflow in which an authorized user provides an identity document to the system. The system processes the document, extracts relevant information, performs the applicable verification steps, and presents the screening information to the user.

A typical demonstration can include:

1. Opening the DISHA AI application.
2. Uploading or providing an identity document.
3. Processing the document using OCR.
4. Extracting relevant identity information.
5. Performing document and consistency checks.
6. Applying available tampering-detection techniques.
7. Performing face verification when applicable.
8. Combining the available verification signals.
9. Displaying the screening result and relevant risk information.
10. Allowing the authorized user to review the result.

The demonstration illustrates the intended end-to-end workflow of DISHA AI and shows how AI-assisted screening can support human verification.

5.4 Limitations

Although DISHA AI provides an AI-assisted framework for identity and document screening, several limitations need to be considered.

- The reliability of document analysis depends on the quality and clarity of the input document.
- Poor-quality, blurred, rotated, or partially damaged documents may reduce OCR performance.
- Different identity documents may have different layouts, formats, security features, and information fields.
- Tampering detection models may require suitable training and validation datasets to achieve

reliable performance.

- Face verification can be affected by image quality, lighting, pose, occlusion, and other environmental conditions.
- AI-generated risk indications should not automatically be treated as proof of identity fraud.
- External databases or authoritative verification services may be required for stronger identity validation.
- The system requires appropriate privacy, access-control, and security mechanisms because identity documents contain sensitive information.
- Model performance can vary depending on the dataset used for training and testing.
- False positives and false negatives remain possible in automated screening systems.

Therefore, DISHA AI should be considered an **AI-assisted screening system**, with final verification decisions remaining under appropriate human and organizational procedures.

5.5 Future Work

The DISHA AI project can be further improved through the following developments:

1. **Larger and Diverse Datasets:** Develop or use appropriately governed datasets containing different document formats and realistic variations.
2. **Improved Tampering Detection:** Incorporate advanced computer-vision and deep-learning techniques for detecting document alterations.
3. **Advanced OCR:** Support multilingual and more complex document layouts to improve information extraction.
4. **Enhanced Face Verification:** Improve face matching under different lighting, pose, and image-quality conditions.
5. **Authoritative Verification Integration:** Where legally and technically permitted, integrate trusted verification services or databases.
6. **Advanced RAG System:** Build a controlled knowledge base containing approved document requirements, verification rules, and reference information.
7. **Agentic Workflow:** Develop specialized AI agents for document analysis, OCR validation, consistency checking, risk analysis, and report generation.
8. **Security Improvements:** Introduce stronger authentication, authorization, encryption, audit logs, and secure storage mechanisms.
9. **Real-Time Screening:** Optimize the system for faster processing in high-volume verification environments.
10. **Performance Evaluation:** Conduct systematic testing using properly prepared datasets and

report metrics such as precision, recall, F1-score, false-positive rate, false-negative rate, and processing time.

11. **User Interface Improvements:** Develop a more comprehensive dashboard that allows authorized users to view extracted information, verification signals, risk indicators, and reports in a clear manner.

5.6 Conclusion

DISHA AI proposes an AI-based approach for assisting with fake identity and document screening. The project combines multiple technologies, including AI, Machine Learning, Deep Learning, NLP, Computer Vision, RAG, and Agentic AI, to create a structured verification workflow.

The proposed system focuses on reducing dependence on manual document examination by assisting with document scanning, OCR-based information extraction, verification, tampering analysis, face verification, and risk-based screening. The combination of these capabilities can provide authorized users with multiple verification signals instead of relying on a single check.

At the same time, automated screening has inherent limitations. The quality of the input document, availability of reliable reference information, model performance, and security of identity data can all influence the system's effectiveness. Therefore, the output of DISHA AI should support—not replace—appropriate human review and established verification procedures.

In conclusion, DISHA AI provides a foundation for developing an intelligent document-screening platform that can be expanded with improved datasets, stronger AI models, secure verification services, RAG-based knowledge retrieval, and agentic workflows. With further development and systematic validation, the proposed framework can serve as a useful foundation for AI-assisted identity and document verification applications.

REFERENCES

1. Muñoz-Haro, J., Tolosana, R., Fierrez, J., Vera-Rodriguez, R., & Morales, A. (2026). “Privacy-aware detection of fake identity documents: methodology, benchmark, and improved algorithms (FakeIDet2).” *Information Fusion*, 128, 103969. DOI: 10.1016/j.inffus.2025.103969.
2. Sirajudeen, M., Anitha, R., Varadarajan, V., Kommers, P., Piuri, V., & Subramaniaswamy,

- V. (2020). "Forgery document detection in information management system using cognitive techniques." *Journal of Intelligent & Fuzzy Systems*, 39(6). DOI: 10.3233/JIFS-189128.
3. Savakar, D. G., Hiremath, R., & Ghuli, A. (2026). "Identity Document Forgery Detection Using Haralick Texture Features." *Pattern Recognition and Image Analysis*, 35, 808–818. DOI: 10.1134/S1054661825700609.
4. Lugon Moulin, S., Weyermann, C., & Baechler, S. (2022). "An efficient method to detect series of fraudulent identity documents based on digitised forensic data." *Science & Justice*, 62(5), 610–620. DOI: 10.1016/j.scijus.2022.09.003.
5. Carneiro, L. A. B. de C. (2026). "Recognition of counterfeit identity documents using FTIR spectroscopy and machine learning approaches." *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 353, 127559. DOI: 10.1016/j.saa.2026.127559.
6. Korshunov, P., Vidit, V., Mohammadi, A., Ecabert, C., Shamoska, N., Marcel, S., et al. (2025). "DeepID Challenge of Detecting Synthetic Manipulations in ID Documents." *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 521–530.
7. Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). "ArcFace: Additive Angular Margin Loss for Deep Face Recognition." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 4690–4699.
8. Schroff, F., Kalenichenko, D., & Philbin, J. (2015). "FaceNet: A Unified Embedding for Face Recognition and Clustering." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 815–823.
9. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., et al. (2020). "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." *Advances in Neural Information Processing Systems (NeurIPS)*, 33.
10. Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., et al. (2024). "A Survey on Large Language Model Based Autonomous Agents." *Frontiers of Computer Science*, 18. DOI: 10.1007/s11704-024-40231-1.
11. Plaat, A., van Duijn, M., van Stein, N., Preuss, M., van der Putten, P., & Batenburg, K. J. (2025). "Agentic Large Language Models, a Survey." *Journal of Artificial Intelligence Research (JAIR)*. DOI: 10.1613/JAIR.1.18675.
12. Kubam, C. S. (2026). "Agentic AI Microservice Framework for Deepfake and Document Fraud Detection in KYC Pipelines." *arXiv preprint*, arXiv:2601.06241

APPENDICES

Technologies Used / Proposed:

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning (DL)
- Natural Language Processing (NLP)
- Computer Vision
- Optical Character Recognition (OCR)
- Retrieval-Augmented Generation (RAG)
- Agentic AI
- Web Technologies
- Git and GitHub

APPENDIX B — SYSTEM WORKFLOW

The overall DISHA AI workflow consists of multiple stages that work together to support identity and document screening.

B.1 Input

The authorized user provides an identity document to the DISHA AI system. The input may be provided in the form of a digital image or scanned document.

B.2 Document Processing

The system preprocesses the uploaded document to improve its suitability for further analysis. Image-processing operations may include resizing, noise reduction, orientation correction, and image-quality assessment.

B.3 OCR Processing

OCR technology is used to extract text from the document. Important fields such as name, document number, date of birth, and other relevant information can be identified from the extracted text.

B.4 Document Verification

The extracted information is checked against predefined rules and available reference information. The system can identify missing fields, inconsistent information, and suspicious patterns.

B.5 Tampering Detection

Computer Vision and Deep Learning techniques can be applied to identify possible alterations in the document. Examples include suspicious text regions, inconsistent image regions, unusual formatting, and other visual anomalies.

B.6 Face Verification

When the document contains a photograph and an appropriate reference image is available, face-verification techniques can be used to compare the faces.

B.7 Risk Analysis

The available verification signals are combined to produce a screening or risk indication. The risk indication is intended to assist an authorized reviewer rather than automatically determine whether an individual has committed fraud.

B.8 Result

The system presents the extracted information, verification findings, detected anomalies, and available risk information in an understandable format.

C.1 Sample Processing

The system performs the following operations:

1. Document upload
2. Image preprocessing
3. OCR-based text extraction
4. Field identification
5. Document consistency checking
6. Tampering analysis
7. Face verification, where applicable
8. Risk-signal generation
9. Result presentation

C.2 Sample Output

Verification Component Result

Document Upload	Processed
OCR Extraction	Completed
Required Fields	Checked
Document Consistency	Checked
Tampering Analysis	Screening completed
Face Verification	Performed where applicable
Risk Analysis	Generated
Final Review	Requires authorized review

APPENDIX D — FUNCTIONAL REQUIREMENTS

The major functional requirements of DISHA AI are listed below.

ID	Requirement
FR-01	The system shall allow an authorized user to provide an identity document.
FR-02	The system shall process the uploaded document image.
FR-03	The system shall extract text using OCR.
FR-04	The system shall identify relevant document fields.
FR-05	The system shall perform document consistency checks.
FR-06	The system shall support document-tampering analysis.
FR-07	The system shall support face verification where applicable.
FR-08	The system shall combine available verification signals.
FR-09	The system shall present screening information to the authorized user.
FR-10	The system shall support secure storage and management of verification records.

APPENDIX E — NON-FUNCTIONAL REQUIREMENTS

E.1 Security

Identity documents contain sensitive information. Therefore, the system should provide appropriate authentication, authorization, secure data handling, and access control.

E.2 Performance

The system should process documents within a reasonable amount of time so that it can support practical screening workflows.

E.3 Scalability

The architecture should allow additional document types, verification modules, and AI models to be incorporated in future versions.

E.4 Usability

The user interface should provide clear instructions and understandable verification results.

E.5 Reliability

The system should handle invalid, incomplete, or poor-quality input appropriately and should avoid presenting uncertain AI outputs as confirmed facts.

E.6 Maintainability

The modular design should allow individual components such as OCR, face verification, tampering detection, RAG, and Agentic AI workflows to be updated independently.

APPENDIX F — TEST CASES

Test Case Input		Expected Result
TC-01	Valid document image	Document is accepted for processing
TC-02	Blurred document	System identifies poor image quality
TC-03	Document without readable text	OCR extraction indicates insufficient text
TC-04	Document with missing fields	Missing information is identified
TC-05	Suspicious document image	Document is flagged for further review
TC-06	Document with photograph	Face-verification module can be invoked
TC-07	Unsupported document format	Appropriate validation message is displayed
TC-08	Empty upload	System requests a valid document
TC-09	Multiple verification inconsistencies	Risk indication is generated
TC-10	Normal document	Screening result is presented for authorized review

APPENDIX G — TECHNOLOGY STACK

Technology	Purpose
Artificial Intelligence	Overall intelligent screening
Machine Learning	Classification and risk analysis
Deep Learning	Image and facial analysis
NLP	Text processing and interpretation
OCR	Extracting text from documents
Computer Vision	Document-image analysis
RAG	Retrieval of information from trusted knowledge sources
Agentic AI	Coordination of multiple AI-based tasks
HTML	Web-page structure
CSS	User-interface styling

Technology	Purpose
JavaScript	Client-side functionality
Git	Version control
GitHub	Source-code repository